

Answer **all** the questions.

1. Find the value of the constant c such that the line $y = 2x + c$ is a tangent to the curve $y = x^2 + 3x + 1$. [3]

2. Express $\frac{18 + 11x - 2x^2}{(x - 1)(x + 2)^2}$ in partial fractions. [5]

3. Prove that $\frac{\cos^2 \theta}{(\operatorname{cosec} \theta - 1)(\operatorname{cosec} \theta + 1)} + \frac{\sin^2 \theta}{(\sec \theta - 1)(\sec \theta + 1)} = 1.$ [5]

4. (a) Find $\frac{d}{dx}(xe^{-2x})$.

[3]

(b) Hence find $\int xe^{-2x} dx$.

[4]

5. Solve the equation $\frac{\cos \theta + 4 \sin \theta}{2 \cos \theta + \sin \theta} = \cot \theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. [6]

6. The function f is given by $f(x) = \frac{ax^2}{x-a}$, for $x > a$, where a is a positive constant.
- (a) Find $f'(x)$.

[2]

The function g , defined for $x > a$, has the property that $g'(x) = (x-a)^2 f'(x)$.
 g decreases for $a < x < 8$.

- (b) Find the value of a .

[4]

7. Find the set of values of the constant k for which the curve $y = kx^2 + 4x + k - 3$ lies completely below the x -axis. [6]

8. The line $y - 2x = 12$ intersects the curve $x^2 - xy + y^2 = 63$ at two points.
Find the coordinates of these two points.

[5]

9. A curve has equation $y = 1 + x - \frac{x^2}{2} - 2x^3$.

(a) Find the x -coordinates of the stationary points, A and B , on the curve.

[4]

It is given that P is a point on the curve where the **gradient is a maximum**.

(b) Show that P and the midpoint of AB have the same x -coordinate.

[4]

10. A circle, with centre C , has equation $x^2 + y^2 + 10x - 24y = 0$.

(a) Find the coordinates of C and the radius of the circle.

[4]

(b) Find the coordinates of the points at which the circle intersects the y -axis.

[2]

The point X is on the line which passes through C and the origin, O . It is given that the distance CX is three times the distance OC .

(c) Find the coordinates of the possible positions of X .

[4]

11. A circular patch of water of negligible thickness is expanding on a thin piece of paper.

At time t seconds the rate of change of the radius, r cm, of the patch is given by

$$\frac{dr}{dt} = \frac{k}{2t+1} \text{ cm/s, where } k \text{ is a constant.}$$

Initially the radius of the patch is 1 cm and the radius is increasing at a rate of 0.5 cm/s.

- (a) Show that $k = 0.5$.

[1]

- (b) Find an expression for r in terms of t .

[4]

- (c) Hence find the rate of increase of the area of the patch after the patch has been expanding for 3 seconds. [4]

12. A ball is thrown vertically upwards. Its height h m, above the ground at time t seconds after being thrown is given by the formula $h = 1.75 + 5t - 5t^2$.

(a) State the height above the ground from which the ball is thrown.

[1]

(b) Express h in the form $a + b(t + c)^2$ where a , b and c are constants to be determined.

[3]

(c) Hence state the maximum height attained by the ball and the time at which this occurs.

[2]

- (d) The ball hits the ground. Explain why the time taken for the ball to hit the ground is **not** twice the time found in part (c). [1]

- (e) Find the length of time for which the ball is at least 2 m above the ground. [3]

13. The perpendicular bisector of the line joining the points $A(2, h)$ and $B(-8, -7)$ passes through the point $X\left(h, -\frac{13}{2}\right)$, where h is a constant.
- (a) Express the gradient of the perpendicular bisector of the line AB in terms of h . [3]

- (b) Hence find the possible values of h . [7]